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Question 1 [C01] [10 Points]

A football jersey shop is running a special promotion for the FIFA World Cup 2026 for fans of Argentina and Germany. Write a Java program to manage the promotion. First, ask the customer to enter the number of jerseys they want to buy. Each jersey costs **Tk. 150**. If the number of jerseys is **0 or less**, display **"Invalid number"** and end the program. Otherwise, ask the customer to choose a team by entering **1 for Argentina** or **2 for Germany**. If an invalid option is entered, display **"No team exists"** and end the program. Next, ask the customer to enter their **date of birth in DDMMYYYY format**. Extract the **first digit of the day** and the **last two digits of the year**, then generate a lucky number as follows:

Argentina: $(\text{first digit} \times 9) + (\text{last two digits} - 5)$

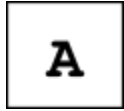
Germany: $(\text{first digit} \times 7) + (\text{last two digits} - 4)$

If the lucky number is **divisible by 7 but not by 3**, apply a **23% discount** to the total bill and display **"Discount Applied"** followed by the final amount to be paid. Otherwise, display **"No Discount Applied"** followed by the total amount to be paid.

Sample Input	Sample Output	Explanation
Enter number of jerseys: 4 Enter team (1-Argentina, 2-Germany): 1 Enter date of birth (DDMMYYYY): 16082017	No Discount Applied Need to pay Tk. 600.0	First digit of the day = 1 Last two digits of the year = 17 Team = Argentina Lucky number = 21 , which is divisible by both 3 and 7 , so the customer does not receive the discount.
Enter number of jerseys: 3 Enter team (1-Argentina, 2-Germany): 2 Enter date of birth (DDMMYYYY): 25062018	Discount Applied Need to pay Tk. 346.5	First digit of the day = 2 Last two digits of the year = 18 Team = Germany Lucky number = $(2 \times 7) + (18 - 4) = \mathbf{28}$, which is divisible by 7 but not by 3 , so the customer receives a 23% discount .
Enter number of jerseys: 2 Enter team (1-Argentina, 2-Germany): 3	No team exists	
Enter number of jerseys: -1	Invalid number	



CSE110: Programming Language I (Lab)
Summer 2026
Lab Exam - 01
Duration: 25 Minutes



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Question 1 [C01] [10 Points]

A sports shop is running a special promotion for the FIFA World Cup 2026 on the flags of Brazil and Portugal. Write a Java program to manage the promotion. First, ask the customer to enter the number of flags they want to buy. Each flag costs **Tk. 55**. If the number of flags is **0 or less**, display **"Invalid number"** and end the program. Otherwise, ask the customer to choose a team by entering **1 for Brazil** or **2 for Portugal**. If an invalid option is entered, display **"No team exists"** and end the program. Next, ask the customer to enter their **date of birth in DDMMYYYY format**. Extract the **first two digits of the day** and the **last digit of the year**, then generate a lucky number as follows:

Brazil: (last digit - 6) + (first two digits × 5)

Portugal: (last digit - 7) + (first two digits × 8)

If the lucky number is **divisible by 4 but not by 8**, apply a **26% discount** to the total bill and display **"Discount Applied"** followed by the final amount to be paid. Otherwise, display **"No Discount Applied"** followed by the total amount to be paid.

Sample Input	Sample Output	Explanation
Enter number of flags: 6 Enter team (1-Brazil, 2-Portugal): 2 Enter date of birth (DDMMYYYY): 24042013	Discount Applied Need to pay Tk. 244.2	First two digits of the day = 24 . Last digit of the year = 3 . Team = Portugal . Lucky number = (3-7)+(24×8) = 188 , which is divisible by 4 but not by 8 , so the customer receives a 26% discount.
Enter number of flags: 5 Enter team (1-Brazil, 2-Portugal): 1 Enter date of birth (DDMMYYYY): 18072015	No Discount Applied Need to pay Tk. 275.0	First two digits of the day = 18 Last digit of the year = 5 Team = Brazil Lucky number = 89 , which is not divisible by 4 , so the customer does not receive the discount.
Enter number of flags: 7 Enter team (1-Brazil, 2-Portugal): 3	No team exists	
Enter number of flags: 0	Invalid number	

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Question 1 [C01] [10 Points]

A space engineer is testing a new rocket by completing **N** launch stages. After completing the **n-th** stage, the rocket earns thrust points equal to **n!** (**n factorial**). If the thrust points are **divisible by both 2 and 3**, they are added to the **Main Fuel Reserve**. Otherwise, if they are **divisible by 2 but not by 3**, they are added to the **Backup Fuel Reserve**. All other thrust points are discarded. However, if the thrust points **exceed 100**, the engine reaches its safety limit. The engineer records the thrust points, prints **"Engine safety limit reached!"**, and immediately stops the test.

Write a Java program to simulate the rocket test. If the test stops because of the safety limit, print the thrust points at which it occurred and the warning message. Otherwise, after all launch stages are completed, print the total points stored in the **Main Fuel Reserve** and the **Backup Fuel Reserve**.

Factorial: $1! = 1$, $2! = 2 \times 1! = 2$, $3! = 3 \times 2! = 6$, $4! = 4 \times 3! = 24$

Sample Input	Sample Output
Enter number of launch stages: 4	Main Fuel Reserve = 30 Backup Fuel Reserve = 2 Explanation: 1! = 1 → Discarded 2! = 2 → Even but not divisible by 3 → Backup Fuel = 2 3! = 6 → Divisible by both 2 and 3 → Main Fuel = 6 4! = 24 → Divisible by both 2 and 3 → Main Fuel = 30
Enter number of launch stages: 8	Thrust Points: 120 Engine safety limit reached! Explanation: 1! = 1 → Discarded 2! = 2 → Backup Fuel 3! = 6 → Main Fuel 4! = 24 → Main Fuel 5! = 120 → Thrust Points exceed 100 , test stops

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Question 1 [C01] [10 Points]

A robotics engineer is testing a new assembly robot by completing **N** production stages. After completing the **n-th** stage, the robot produces energy units equal to **n!** (**n factorial**). If the energy units are **divisible by 4**, they are stored in the **Primary Battery**. Otherwise, if they are **even but not divisible by 4**, they are stored in the **Secondary Battery**. All other energy units are discarded. However, the robot has an overheating limit. If the energy units become **greater than 500**, the engineer records the energy units, prints "**Robot overheated!**", and immediately stops the test.

Write a Java program to simulate the robot test. If the test stops because of overheating, print the energy units at which it occurred and the warning message. Otherwise, after all production stages are completed, print the total energy stored in the **Primary Battery** and the **Secondary Battery**.

Factorial: $1! = 1$, $2! = 2 \times 1! = 2$, $3! = 3 \times 2! = 6$, $4! = 4 \times 3! = 24$

Sample Input	Sample Output
Enter number of production stages: 4	Primary Battery = 24 Secondary Battery = 8 Explanation: 1! = 1 → Discarded 2! = 2 → Even but not divisible by 4 → Secondary Battery = 2 3! = 6 → Even but not divisible by 4 → Secondary Battery = 8 4! = 24 → Divisible by 4 → Primary Battery = 24
Enter number of production stages: 7	Energy Units: 720 Robot overheated! Explanation: 1! = 1 → Discarded 2! = 2 → Secondary Battery 3! = 6 → Secondary Battery 4! = 24 → Primary Battery 5! = 120 → Primary Battery 6! = 720 → Energy units exceed 500 , so the test stops immediately.

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Question 1 [C01] [10 Points]

A hospital has multiple emergency wards. The first ward contains **1** patient bed, the second ward contains **2** patient beds, the third ward contains **3** patient beds, and so on. A nurse visits each bed to record the patient's body temperature, inspecting the wards one by one. A temperature **below 30°C or above 45°C** is considered invalid. In such cases, the nurse immediately takes the patient's temperature again. Invalid readings should not be included in any calculation. If a hospital-wide emergency occurs during the inspection, the nurse enters **-1 as a special input instead of a temperature**. Once **-1** is entered, the inspection is terminated immediately, and no further temperatures are recorded.

Write a Java program that **first takes the number of wards as input** and then records the temperature of every patient according to the rules above. After all inspections, display the **ward number with the highest number of patients with fever** (temperature **38°C or above**). If multiple wards have the same highest number of fever patients, display the **first such ward**. If the inspection is terminated due to an emergency, display the **ward number and bed number** where the emergency occurred.

Sample Input 1	Sample Input 2	Sample Input 3
Enter the number of wards: 3 Enter temperature(s) for :- Ward 1: 37 Ward 2: Bed 1: 40 Bed 2: 38 Ward 3: Bed 1: 38 Bed 2: 40 Bed 3: 37	Enter the number of wards: 3 Enter temperature(s) for :- Ward 1: Bed 1: 37 Ward 2: Bed 1: 46 Bed 1: 39 Bed 2: 36 Ward 3: Bed 1: 38 Bed 2: 40 Bed 3: 41	Enter the number of wards: 4 Enter temperature(s) for :- Ward 1: Bed 1: 36 Ward 2: Bed 1: 38 Bed 2: 37 Ward 3: Bed 1: 41 Bed 2: -1
Sample Output 1	Sample Output 2	Sample Output 3
Ward 2 has the highest number of patients with fever.	Invalid temperature! Enter temperature again for Bed 1 at Ward 2 Ward 3 has the highest number of patients with fever.	Emergency occurred at Ward 3, Bed 2 Ward 2 has the highest number of patients with fever.

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Question 1 [C01] [10 Points]

An airport has multiple security checkpoints. The first checkpoint screens **1** passenger, the second checkpoint screens **2** passengers, the third checkpoint screens **3** passengers, and so on. A security officer checks each passenger one by one to record the weight of their carry-on baggage. A baggage weight **below 0 kg or above 15 kg** is considered an invalid reading. In such cases, the officer immediately measures the baggage again. Invalid readings should not be included in any calculation. If an airport-wide emergency occurs during the screening process, the officer enters **-1** as a special input instead of a baggage weight. Once **-1** is entered, the screening process is terminated immediately, and no further passengers are screened.

Write a Java program that first takes the number of security checkpoints as input and then records the baggage weight of every passenger according to the rules above. After all passengers have been screened, display the **checkpoint number with the highest number of passengers carrying heavy baggage** (baggage weight **10 kg or above**). If multiple checkpoints have the same highest number of heavy-baggage passengers, display the **first such checkpoint**. If the screening process is terminated due to an emergency, display the **checkpoint number and passenger number** where the emergency occurred.

Sample Input 1	Sample Input 2	Sample Input 3
Enter the number of checkpoints: 3 Enter weight(s) for:- Checkpoint 1: Passenger 1: 8 Checkpoint 2: Passenger 1: 10 Passenger 2: 12 Checkpoint 3: Passenger 1: 9 Passenger 2: 11 Passenger 3: 10	Enter the number of checkpoints: 3 Enter weight(s) for:- Checkpoint 1: Passenger 1: 7 Checkpoint 2: Passenger 1: 18 Passenger 1: 11 Passenger 2: 9 Checkpoint 3: Passenger 1: 10 Passenger 2: 12 Passenger 3: 13	Enter the number of checkpoints: 4 Enter weight(s) for:- Checkpoint 1: Passenger 1: 8 Checkpoint 2: Passenger 1: 10 Passenger 2: 9 Checkpoint 3: Passenger 1: 11 Passenger 2: -1
Sample Output 1	Sample Output 2	Sample Output 3
Checkpoint 2 has the highest number of passengers with heavy baggage.	Invalid baggage weight! Enter weight again for Passenger 1 at Checkpoint 2 Checkpoint 3 has the highest number of passengers with heavy baggage.	Emergency occurred at Checkpoint 3, Passenger 2 Checkpoint 2 has the highest number of passengers with heavy baggage.

SET - [A]

SOLUTION

```

import java.util.Scanner;
public class Exam3SetA {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of wards: ");
        int wards = sc.nextInt();
        int maxFever = -1;
        int wardWithMaxFever = 1;
        boolean emergency = false;
        System.out.println("Enter temperature(s) for:- ");
        for (int ward = 1; ward <= wards; ward++) {
            System.out.println("Ward " + ward + ":");
            int feverCount = 0;
            for (int bed = 1; bed <= ward; bed++) {
                System.out.print("Bed " + bed + ": ");
                int temp = sc.nextInt();
                if (temp == -1) {
                    emergency = true;
                    System.out.println("Emergency occurred at Ward " + ward + ", Bed " + bed);
                    break;
                }
                if (temp < 30 || temp > 45) {
                    System.out.println("Invalid temperature! Enter temperature again for Bed "
                        + bed + " at Ward " + ward);
                    bed -= 1;
                    continue;
                }
                if (temp >= 38) {
                    feverCount++;
                }
            }
            if (emergency==true) {
                break;
            }
            else if ((emergency==false) && (feverCount > maxFever)) {
                maxFever = feverCount;
                wardWithMaxFever = ward;
            }
        }
        System.out.println("Ward "+wardWithMaxFever+" has the highest number of patients with fever.");
    }
}

```

SET - [B]

SOLUTION

```

import java.util.Scanner;
public class Exam3SetB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of checkpoints: ");
        int checkpoints = sc.nextInt();
        int maxHeavy = -1;
        int checkpointWithMaxHeavy = 1;
        boolean emergency = false;
        System.out.println("Enter weight(s) for:- ");
        for (int checkpoint = 1; checkpoint <= checkpoints; checkpoint++) {
            System.out.println("Checkpoint " + checkpoint + ":");
            int heavyCount = 0;
            for (int passenger = 1; passenger <= checkpoint; passenger++) {
                System.out.print("Passenger " + passenger + ": ");
                int weight = sc.nextInt();
                if (weight == -1) {
                    emergency = true;
                    System.out.println("Emergency occurred at Checkpoint " + checkpoint + ", Passenger " + passenger);
                    break;
                }
                if (weight < 0 || weight > 15) {
                    System.out.println("Invalid baggage weight! Enter weight again for Passenger " + passenger + " at Checkpoint " + checkpoint);
                    passenger -= 1;
                    continue;
                }
                if (weight >= 10) {
                    heavyCount++;
                }
            }
            if (emergency==true) {
                break;
            }
            else if ((emergency==false) && (heavyCount > maxHeavy)) {
                maxHeavy = heavyCount;
                checkpointWithMaxHeavy = checkpoint;
            }
        }
        System.out.println("Checkpoint "+checkpointWithMaxHeavy+" has the highest number of passengers with heavy baggage.");
    }
}

```

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Question 1 [C02] [10 Points]

BRAC University wants to encrypt employee codes before storing them in its system. Write a Java program that takes an employee code as input and generates an encrypted version of the code by processing each character according to the following rules:

1. **Lowercase letters (a - z)**
 - a. Shift each **even-index lowercase** letter **forward** by **2 positions**; **odd-index** lowercase letters remain unchanged.
 - b. If the shift goes beyond 'z', wrap around and continue again from 'a'.
2. **Uppercase letters (A - Z)**
 - a. Shift each **odd-index uppercase** letter **backward** by **1 position**; **even-index** uppercase letters will remain unchanged.
 - b. If the shift goes before 'A', wrap around and continue again from 'Z'.
3. **Digits (0 - 9)**
 - a. Replace every **even digit** with *.
 - b. Replace every **odd digit** with #.
4. **Special characters** (such as @, #, \$, %, !, etc): Replace every special character with '- '.

Finally, **print** the encrypted employee code.

Sample Input	Sample Output	Explanation
Enter Employee Code: zyAZ38@!	Encrypted Code: byAY#*--	z → lowercase + even index → + 2 → b y → lowercase + odd index → unchanged → y A → uppercase + even index → unchanged → A Z → uppercase + odd index → -1 → Y 3 → odd digit → # 8 → even digit → * @ → special character → - ! → special character → - Output : byAY#*--

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Question 1 [C02] [10 Points]

Foodpanda wants to encrypt employee codes before storing them in its system. Write a Java program that takes an employee code as input and generates an encrypted version of the code by processing each character according to the following rules:

1. Lowercase letters (a - z)

- Shift each **odd-index lowercase** letter **backward** by **2 positions**; even-index lowercase letters remain unchanged.
- If the shift goes before 'a', wrap around and continue again from 'z'.

2. Uppercase letters (A - Z)

- Shift each **even-index uppercase** letter **forward** by **1 position**; odd-index uppercase letters remain unchanged.
- If the shift goes beyond 'Z', wrap around and continue again from 'A'.

3. Digits (0 - 9) :

- Replace every **even digit** with #.
- Replace every **odd digit** with *.

4. Special characters (such as @, #, \$, %, !): Replace every special character with '='.

Finally, **print** the encrypted employee code.

Sample Input	Sample Output	Explanation
Enter Employee Code: baZY38@!	Encrypted Code: byAY*#==	b → lowercase + even index → unchanged → b a → lowercase + odd index → -2 → y Z → uppercase + even index → +1 → A Y → uppercase + odd index → unchanged → Y 3 → odd digit → * 8 → even digit → # @ → special character → = ! → special character → = Output : byAY*#==

SET - [A]

SOLUTION

```

import java.util.Scanner;

public class CodeEncryption1 {
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter Employee Code : ");
        String code = scanner.nextLine();

        String encryptedCode = "";
        for (int i = 0; i < code.length(); i++) {

            char ch = code.charAt(i);
            // Lowercase letters
            if (ch >= 'a' && ch <= 'z') {

                if (i % 2 == 0) {
                    ch = (char) (ch + 2);

                    if (ch > 'z') {
                        ch = (char) (ch - 26);
                    }
                }
            }
            // Uppercase letters
            else if (ch >= 'A' && ch <= 'Z') {

                if (i % 2 == 1) {
                    ch = (char) (ch - 1);

                    if (ch < 'A') {
                        ch = 'Z';
                    }
                }
            }

            // Digits
            else if (ch >= '0' && ch <= '9') {

                int digit = ch - '0';

                if (digit % 2 == 0) {
                    ch = '*';
                } else {
                    ch = '#';
                }
            }

            // Special characters
            else {
                ch = '-';
            }

            encryptedCode = encryptedCode + ch;
        }
        System.out.println("Encrypted Code : " + encryptedCode);
    }
}

```

SOLUTION

```
import java.util.Scanner;

public class CodeEncryption2 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter Employee Code : ");
        String code = scanner.nextLine();
        String encryptedCode = "";
        for (int i = 0; i < code.length(); i++) {
            char ch = code.charAt(i);
            // Lowercase letters
            if (ch >= 'a' && ch <= 'z') {
                if (i % 2 == 1) {
                    ch = (char) (ch - 2);

                    if (ch < 'a') {
                        ch = (char) (ch + 26);
                    }
                }

                // Uppercase letters
            } else if (ch >= 'A' && ch <= 'Z') {
                if (i % 2 == 0) {
                    ch = (char) (ch + 1);
                    if (ch > 'Z') {
                        ch = 'A';
                    }
                }
            }

            // Digits
            else if (ch >= '0' && ch <= '9') {
                int digit = ch - '0';

                if (digit % 2 == 0) {
                    ch = '#';
                } else {
                    ch = '*';
                }
            }

            // Special characters
            else {
                ch = '=';
            }

            encryptedCode = encryptedCode + ch;
        }
        System.out.println("Encrypted Code : " + encryptedCode);
    }
}
```